**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

**MG4DSEPHYA00** **Physics of Medical Diagnostics**

Duration: 2 hrs Maximum Marks: 50

*Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create (C), Skill(S), Interest(I) and Appreciation(An)*

*Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.*

**Part A**

Short Answer Type

Questions Answer any **10**

questions Each question

carries **2** Mark

1 What do you meant by spatial resolution in medical imaging? [K][CO1]
2 What are the factors affecting attenuation of an image data?[U][CO1]
3 Name two applications of imaging sensor technology. [U][CO1]
4 What are the different parts of a X ray machine?[U][CO2]
5 What is the basic principle of MRI technique? [U][CO4]
6 Define PET imaging.[K][CO5]
7 Mention any two applications of ultrasound in medical diagnosis.[K][CO3]
8 What are the factors affecting image quality? [K][CO1]
9 Define motion mode ultrasound technique.[U][CO5]
10 What do you mean by chemical shift in NMR? [U][CO4]
11 What are the various Radiation Protection measures in Nuclear Medicine? [U][CO5]
12 What is the peculiarity of 6th generation CT scanners? [K][CO4]
13 Define B-scan ultrasonography [U][CO5]
14 What is the principle behind mammography?[U][CO2]

(10x2=2)

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

15 Differentiate between PET and SPECT imaging [K][CO2]
16 Explain how data acquisition and image reconstruction are done in medical imaging.[A][CO1]
17 Describe the Generations of CT Scanners[U][CO4]
18 Explain how maintenance of X-ray equipment is done in hospitals.[K][CO3]
19 Differentiate between fluoroscopy and mammography. [A][CO2]
20 What is the basic instrumentation in MRI scanner? [U][CO2]
21 Explain the algorithm for tomography[U][CO4]

22 Explain the colour doppler technique in ultrasound imaging .[U][CO5]

(6x5=30)